Problem Set 1

Due Online: Thursday, February 6, 11:59 pm

1. Derive the autocorrelation function of

$$y_t = 1 + \varepsilon_t + 0.6\varepsilon_{t-1} + 0.2\varepsilon_{t-2},$$

where $\varepsilon_t \sim WN(0,1)$, i.e., a white noise process with zero mean and unit variance. Show your work.

2. Consider the time series

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t + \theta\varepsilon_{t-1},$$

where $\varepsilon_t \sim WN\left(0, \sigma^2\right)$, and β_0, β_1, and θ are fixed constants ($\beta_1 \neq 0$).

   (a) Determine whether y_t is weakly stationary.

   (b) Determine whether $x_t = y_t - y_{t-1}$ is weakly stationary.

3. Consider the $MA(1)$ process:

$$y_t = \varepsilon_t + \theta\varepsilon_{t-1}, \quad \varepsilon_t \sim WN\left(0, \sigma^2\right).$$

   (a) Suppose $|\theta| < 1$. Show that ε_t can be expressed in terms of y_s, $s \leq t$. Do more recent observations have higher weights than observations from the distant past?

   (b) Suppose now $|\theta| > 1$. Show that ε_t can be expressed in terms of y_s, $s \geq t$.
   Hint: Instead of doing backward substitution, use forward substitution starting from $\varepsilon_t = \theta^{-1}\varepsilon_{t+1} + \theta^{-1}y_{t+1}$.
   Comment: Note that in this case the series you obtained in (a) does not converge, and it indicates that more distant observations have a greater influence on the current error.

4. Let

$$x_t = \varepsilon_t - 0.4\varepsilon_{t-1}, \quad \varepsilon_t \sim WN(0,4),$$
$$y_t = u_t - 0.7u_{t-1}, \quad u_t \sim WN(0,1),$$

where $\{\varepsilon_t\}$ is distributed independently of $\{u_t\}$. Consider $z_t = x_t + y_t$.

   (a) Find the autocovariances of $\{z_t\}$.

(b) Express z_t as an invertible MA process.
Hint: Suppose that $z_t = e_t + \theta e_{t-1}$, where $e_t \sim WN\left(0, \sigma_e^2\right)$. You will need to solve for θ and σ_e^2 such that z_t has the same autocovariances obtained in part (a).

5. Consider the following $AR(2)$ process:

$$y_t = y_{t-1} - 0.24y_{t-2} + \varepsilon_t, \quad \varepsilon_t \sim WN\left(0, \sigma^2\right).$$

(a) Express this process using the lag operator L.

(b) Is this process weakly stationary? Does y_t exhibit stochastic business cycles?

(c) Find the coefficients $\{\psi_i\}$ in the MA representation $y_t = \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i}$.

(d) Consider now

$$x_t = 2 + x_{t-1} - 0.24x_{t-2} + \varepsilon_t.$$

Find the MA representation of x_t. (*Hint*: First find the mean of x_t and rewrite this process as an $AR\,(2)$ with zero mean.)

(e) Check for stationarity and stochastic business cycles for the process

$$z_t = z_{t-1} - 0.3z_{t-2} + \varepsilon_t.$$

Find the average length of the stochastic cycles.

6. (Computer Exercise) In this exercise, let us examine the stock returns of Walmart, Inc. (WMT) from 2010 to 2019. The data can be obtained from Yahoo Finance using the `quantmod` package in R. You need to install the package (`install.packages("quantmod")`) if you are using it for the first time. Once it is installed, you can load it into your current R session by `library(quantmod)`, and then retrieve the data by

`getSymbols("WMT", src = "yahoo", from = "2010 − 01 − 01", to = "2019 − 12 − 31")`

(a) How many variables and observations are in the WMT data?

(b) Consider the daily close prices. You can retrieve the close prices by
`p < −WMT$WMT.Close`
Draw the time series plot of the close prices. Does it look like a stationary series? Plot the ACF of `p` and find the values of the first four lags (i.e., $\rho_1, ..., \rho_4$). Do you find strong serial dependence in the share price?

(c) Generate daily log returns (also known as continuously compounded returns) by
`rd < −diff(log(p))`
and remove missing values. Examine the `rd` series by answering the same questions as in part (b).

(d) Inspect a snippet of your data by printing the first 5 observations of `rd`. (The R function is `head`().) Report the mean, median, and standard deviation of `rd` (over the entire sample period).

(e) Since `rd` is stored as an `xts` object, it is easy to calculate weekly and monthly log returns using the built-in functions

`rw` $<-$ `apply.weekly(rd, FUN = sum)`

`rm` $<-$ `apply.monthly(rd, FUN = sum)`

(Refer to the help documentation for details about the above functions.)

i. How many observations do you have on `rw` and `rm`?

ii. Draw the time series plots of daily, weekly and monthly returns in one figure by arranging them in 3 rows and 1 column. Name the three plots as "daily returns", "weekly returns", and "monthly returns", respectively.

(In R you can use `par(mfrow = c(3, 1))` before `plot`() to combine plots, and use `par(mfrow = c(1, 1))` to restore the plotting arrangement to its original state. You can change the main title of a plot by the `main` option.)

iii. Plot the ACFs of `rw` and `rm` (in separate figures).

iv. Conduct the Ljung–Box tests using 10 lags on the daily, weekly, and monthly returns, respectively. Write down the null and alternative hypotheses. Draw your conclusion.

(f) Comment on the similarities and differences between returns at different frequencies based on your findings in (c) and (e).